Physics-Guided Attack Generation for Resilient Cyber-Physical State Estimation Algorithm Design for Resilient Cyber-Physical Systems using an Automated Attack Generative Model

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***Abstract*—This paper presents a suite of algorithms for detecting and localizing attacks in develops a resilient state-estimation framework for linear cyber-physical systems , and performing improved resilient state estimation through a pruning algorithm. High performance rates for the underlying detection and localization algorithms are achieved by generating training data that cover large region of the attack space. An unsupervised generative model trained by physics-based discriminators is designed to generate successful false data injection attacks. Then the generated adversarial examples are subject to sparse false-data injection attacks. A steady-state Kalman filter and a chi-square innovation detector are incorporated into a physics-guided attack generator. The generator uses the closed-loop attack-to-output map to quantify attack effectiveness and the attack-to-innovation map to quantify detection probability. Generated attacks are subsequently used to train a multi-class deep neural networkwhich detects and localizes the attacks on measurements. Next, a pruning algorithm is included to improve the precision of localization result and provide performance guarantees for the resulting resilient observer. The performance of the proposed method is validated using numerical simulations of a water distribution cyber-physical systemchannel-wise localization network. Under explicit conditional-calibration and independence assumptions, calibrated safety probabilities are employed to select a measurement subset with a prescribed probability of being attack-free. The resulting analysis relates four quantities that are commonly treated separately: attack sparsity, sensor-set observability, localization reliability, and conditional Kalman-filter error covariance. A water-distribution case study reports the empirical observability profile and illustrates the localization improvement obtained from an earlier physics-guided training set; numerical validation of the revised KF/chi-square generator is left for future work.**

***Index Terms*—Resilient Control and Estimation, False Data Injection Attack, Generative Model, Multilayer Perceptron, Pruning algorithm estimation, Kalman filter, chi-square detector, false data injection attack, generative model, multilayer perceptron, pruning, observability**

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I. Introduction

Modern cyber-physical critical infrastructures (CPCI) are facing fast-evolving cyber threats. Cyberattacks can strongly impact the operation of the CPCIs. For instance, a suspected cyber intruder took control of the Prykarpattyaoblenergo power system control center in western Ukraine in December 2015, leaving 230,000 people without electricity for up to 6 hours. That was the first time hackers have successfully targeted a country's power grid [1]. Also, cyber attacks in water systems have already become a reality. In 2015Cyber-physical critical infrastructure relies on networked sensing, estimation, and control. This dependence creates an attack surface through which corrupted measurements can alter both the estimated state and the resulting control action. False-data injection attacks (FDIAs) have been studied extensively in power, water, 25 cyber attacks were disclosed in several water systems [2]. And recently in 2020, a malicious cyber-attack attempted to raise the chlorine level in Israel's watersupply to a dangerous proportion [3]. Supervisory Control and Data Acquisition (SCADA) is a critical part of the CPCI that is highly susceptible to cyberattacks. It is responsible for collecting measurements from Remote Terminal Units (RTU) and sending them to the Control Centers (CC) to perform various functions such as contingency analysis, optimal planning, state estimation (SE), etc. The primary purpose of those functions is to maintain stable and secure operation of the CPCI.

SE is a core function in the CCs which monitors the system states using information collected from the SCADA systems and other measurement devices. SE can also detect and identify bad data that can cause significant error in the resulting estimate. The SE algorithm's accuracy depends on how pure and accurate the system measurements are. If any malicious or erroneous measurements pass through undetected, it could mislead the operators into making catastrophically wrong decisions. Numerous studies have shown that traditional SE algorithms are vulnerable to False Data Injection Attacks (FDIAs), where an intruder aims to hack multiple RTUs or even communication channels to insert fake measurements in order to misguide the CCs operational process [4]–[7]. Since FDIA was introduced in [4], different researchers have proposed various techniques for detection and localization, including diagnostic robust generalized potential [8], generalized likelihood ratio [9], fast Go-Decomposition (GoDec) [10], Markov chain [11], Bayesian detection with binary hypothesis [12], modified unbiased finite impulse response (UFIR) estimator [13], and cosine similarity matching scheme [14] and some residual-based approaches such as the Kullback Leibler distance method [15], unscented Kalman filter [16], χ^2 failure detector [17] and a residual-based localization scheme [18]. However, due to their dependence on the system model, the associated model uncertainty would affect their accuracy. Thus, if the FDIAs are designed appropriately, these model-based or residual-based detectors can easily be bypassed [4], [18]. Take for example, filter-based attack detector designs [13], [16]; the recovered attack signals could have deviations due to the uncertainties of the underlying model used in the filter design. Thus, if the attacker substitutes current meter measurements with historical readings following the same distribution, the Bayesian framework and generalized likelihood ratio

approaches are unable to identify this FDIA [19].

To overcome the limitations of traditional residual-based bad data detection approaches, data-driven solutions based on machine learning algorithms have been widely adopted for detecting and localizing FDIAs due to their fast execution times and accurate results [20]–[26]. [20] utilized and tested various machine learning algorithms in detecting FDIAs. The results showed that machine learning algorithms can detect FDIAs accurately and faster. [21] proposed a deep neural network algorithm that can automatically detect and localize FDIAs in power systems. The proposed method can integrate the underlying graph topology of the grid and spatial correlations of its measurement data to jointly detect and localize the FDIAs in power systems. [26] proposed a hierarchy of neural networks with the capability of detecting novel attacks by training networks to understand the entire normal space instead of attack space. and other infrastructure systems [1]–[4]. A particularly important class of FDIAs exploits knowledge of the estimator or detector to produce a significant physical effect while maintaining a small detection statistic [17].

In addition to attack detection and localization, resilient estimation approaches focus on maintaining the correctness of state estimates in the presence of adversarial attacks, which enables resilient operation of the CPSs [27]. Although attacks are possibly unbounded and can be designed to "fool" ℓ_2 observers [4], [17], the attack signals are sparse due to the resource limitation of the attackers. Based on these characteristics, the authors in [28] proposed an ℓ_1 minimization program for estimator design on linear systems and proposed an upper-bound on the total number of compromised measurements for guaranteed secure estimation. Some extension work considered robustness on resilient ℓ_1 estimation in presence of modeling error [29], worst-case estimation error bound in presence of bounded noise [30], and $\ell_1 - \ell_2$ estimation scheme [31]. And some iterative resilient observer designs has been proposed for determinant system, such as event-trigger Luenburger Observer [32] and Gramian-based estimator [33]. For stochastic system, several attack-resilient control strategies have also been studied [34], [35]. In addition, resilient estimation and control design has also been studied for nonlinear systems [36]. Model-based bad-data detection commonly evaluates an innovation or residual against a prescribed threshold. Examples include generalized likelihood-ratio tests, Bayesian detectors, and chi-square tests associated with Kalman filtering [9], [12], [17]. These methods offer interpretable statistical thresholds, but their performance depends on the plant and noise models. Data-driven classifiers provide a complementary mechanism for identifying compromised channels and have been applied to FDIA detection and localization [20]–[22]. Their effectiveness, however, depends strongly on whether the training data represent attacks near the decision boundary.

Furthermore, some enhanced resilient CPS designs have been studied to accommodate even worse adversarial condition. To push the limit on the number of compromised sensors below which secure estimation would be guaranteed, additional prior information could be used in the resilient estimation design. Existing results have used such information as measurement prior, support prior, and state prior: In [37], three forms of *state prior* are considered: *sparsity information*, $(\alpha, \overline{n}_0)$ *sparsity information,*, and *side information*. These are information generally unavailable to the malicious agent. In [35], the authors designed an improved resilient estimator, with input and state constraints, against actuator attacks. In our previous work [38]–[40], the resiliency of ℓ_1 based observers is enhanced by utilizing a measurement prior, generated by a pre-trained Gaussian processAdversarial data generation addresses this limitation by enriching the training distribution. Existing approaches include optimization-based attack synthesis and generative models trained from attack corpora [41]–[43]. The former can be computationally demanding, whereas the latter presuppose a representative collection of feasible attacks. In [44], a resilient unscented Kalman filter was proposed based on improved support prior. In [18], a weighted A physics-guided generator provides an alternative: known plant and estimator dynamics can be used as fixed discriminators, allowing attack samples to be generated without first collecting a labeled attack dataset.

Attack localization alone does not ensure resilient estimation. A localization model may retain compromised measurements or discard measurements that are important for observability. Secure and resilient estimators therefore exploit attack sparsity, sensor redundancy, and support information [18], [28]–[30]. Prior work has investigated ℓ_1 observer is designed to exploit the support prior more rigorouslyestimators, support-weighted estimators, and resilient Kalman filters [31], [44]. A remaining design question is how localization reliability should be translated into a sensor subset whose cleanliness, observability, and estimation error can be quantified simultaneously.

Alternatively, improved resiliency of CPS could also be achieved by using more accurate attack detection and localization algorithms. For data-driven attack detectors, especially supervised learning-based detectors, exploiting more of the attack space and system nominal conditions could improve accuracy significantly [45]. This has motivated researchers to study attack data generation strategies for improving the detection/localization boundaries of the underlying data-driven algorithms. The goal of the attack generation problem is to generate feasible attacks that could elicit significant targeted disruption on system, while bypassing bad data detection. Model-based attack generation strategies often assume the attackers have the knowledge of system models [4], [17]. This assumption limits the application range of those attack generation strategies. Thus, automated attack generation without knowledge of the system models, is currently gaining a lot of attention. Although this topic has been studied widely in the machine learning area, for example; adversarial machine learning [46] and deepfake [47], the automated attack generation problem for CPS is a new challenging problem since the physical process dynamics is included. Inspired by the generative adversarial network (GAN) [48], some earlier researches [42], [43], [49] have studied GAN-based attack generation by using an existing attack dataset. The idea is to use GAN to learn the underlying distribution of feasible attacks, then a richer dataset of attacks could be generated. However, it is not practical to assume the existence of a non-generative attack dataset. Thus, an automated attack generation process, without attack examples, is needed. To the best of the authors knowledge, this paper is the first paper to study this problem.

*Contribution*Contributions: This paper proposes an enhanced resilient solution for CPSs subject to FDIAs. It is a hybrid, physics-based and data-driven, detection-based resilient estimation framework. The proposed framework consists of an automated attack generator, supervised learning-based attack detection and localization algorithms, pruning algorithms, and a state estimator. To train a supervised learning-based detector, abundant adversarial attack examples are often required. Model-based attack generators such as [18], [41] generally adopt optimization-based mechanisms which are computationally expensive, even NP-hard most times. Most traditional data-driven attack generators [42], [43], in literature, still require the attack examples with labels for training. In this paper, we propose an unsupervised generative model (GM) trained by multiple addresses that question in a Kalman-filter setting. First, finite-horizon maps are derived from a sparse measurement attack to the closed-loop critical output and to the normalized Kalman-filter innovation. These maps define physics-based discriminators derived from the plant model. Secondly, due to the inherent uncertainty in the precision of data-driven detection and localization approaches, a pruning algorithm is proposed to systematically improve the precision of the resulting localization. The corresponding improvement guarantee is quantified ana-

lytically as a function of the aggressiveness of the pruning algorithm. Consequently, it is shown that the states of the system can be recovered correctly, thereby maintaining the operational performance of the CPS while attack is underway. Finally, the generalization of the proposed resilient design framework is demonstrated by showing how different state estimators and different supervised learning-based detectors could be placed in this frame. for generating attacks subject to an explicit chi-square detection-probability constraint. Second, a channel-wise localization network is combined with a calibrated sensor-selection rule. Under stated conditional-calibration and independence assumptions, the rule guarantees that the particular subset supplied to the estimator is attack-free with a prescribed probability. Third, the analysis relates the realized attack sparsity, the observability threshold, the selection reliability, and the conditional subset-KF covariance. This relation quantifies the estimation penalty incurred when higher selection reliability requires fewer measurements.

The reminder The remainder of the paper is organized as follows: All the notations and the necessary mathematical tools used in the development are given in . Section II . In introduces the notation. Section III , the model of CPS is described. In presents the stochastic CPS model, Kalman filter, and chi-square detector. Section IV , a formal definition of successful FDIAs is presented, and a data-driven generative model is proposed to generate successful FDIAs. In derives the attack-response maps and formulates physics-guided attack generation. Section V , detection and localization approaches based on multi-class MLP are given and a pruning algorithm is included to improve the resulting localization precision. In Section VI, a resilient ℓ_2 observer is designed to maintain correct state estimation under adversarial attacks. Conclusion remarks follow in Section VIII.develops channel-wise localization and calibrated sensor selection. Section VI analyzes the resulting subset Kalman filter. Section VII presents the water-distribution case study, and Section VIII concludes the paper.

II. Notation and Preliminary

The following notations and definitions are used throughout the paper: $\mathbb{R}$, $\mathbb{R}^n$, $\mathbb{R}^{n\times m}$ denote the space The sets of real numbers, real vectors of length n, and real matrices of real $n$rows and m columns respectively. $\mathbb{R}_+$ denotes the space of positive real numbers. Normal-face lower-case letters $(e.g.\ x \in \mathbb{R})$ are used to represent real scalars, bold-face lower-case letters $(e.g.\ \mathbf{x} \in \mathbb{R}^n)$ represent vectors, while normal-face upper-case letters $(e.g.\ X \in \mathbb{R}^{n\times m})$ represent matrices. $X^\top$ denotes the transpose of the -vectors, and real $m \times n$ matrices are denoted by $\mathbb{R}$, $\mathbb{R}^n$, and $\mathbb{R}^{m\times n}$, respectively. For a matrix X. The spectral radiusof the square matrix X , $X^\top$, $X^\dagger$, $\rho(X)$, and $\mathrm{tr}(X)$ denote its transpose, Moore–Penrose inverse, spectral radius, and trace. The identity matrix of dimension n is denoted by $\rho(X)$. $X^\dagger \triangleq \left(X^\top X\right)^{-1} X^\top$ I_n. For symmetric matrices, $X \succeq 0$ and $X \succ 0$ denote positive semidefiniteness and positive definiteness.

For an index set $\mathcal{S}$, $X_{\mathcal{S}}$ denotes the Moore-Penrose inverse. $\mathbf{1}_n$ and I_n denote the vector of ones and the identity matrix of size n respectively. Let $\mathcal{T} \subseteq \{1,\dots,n\}$, then, for a matrix $X \in \mathbb{R}^{m\times n}$, $X_{\mathcal{T}} \in \mathbb{R}^{|\mathcal{T}|\times m}$ is the submatrix obtained by extracting submatrix formed from the rows of X corresponding to the indices in $\mathcal{T}$indexed by $\mathcal{S}$, and $\mathcal{S}^c$ denotes its complement relative to the applicable sensor set. $\mathcal{T}^c$ denotes the complement of a set $\mathcal{T}$, and the universal seton which it is defined will be clear from the context. We use I_T (or simply I when T is clear from context) to represent a T-time window $[i-T+1, i]$. In the same vein, $I-1$ is used to denote the time window $[i-T, i-1]$ accordingly The cardinality of $\mathcal{S}$ is $|\mathcal{S}|$. The support and zero norm of a vector $\mathbf{x} \in \mathbb{R}^n$ is the set of the indices of nonzero entries in $\mathbf{x}$, defined as $\mathrm{supp}(\mathbf{x}) \triangleq \{i \subseteq \{1,\dots,n\} | \mathbf{x}_i \neq 0\}$. A vector $\mathbf{x} \in \mathbf{R}^n$ is said to be are

$$\mathrm{supp}(\mathbf{x}) = \{j : x_j \neq 0\}, \qquad \|\mathbf{x}\|_0 = |\,\mathrm{supp}(\mathbf{x})|.$$

A vector is k-sparse if $|\mathrm{supp}(\mathbf{x})| \leq k$, and Σ_k denotes the subset of k-sparse vectors. A moving-horizon vector $\mathbf{x}_I \in \Sigma_k$ is given by $\mathbf{x}_I = \begin{bmatrix}\mathbf{x}_1^\top & \cdots & \mathbf{x}_T^\top\end{bmatrix} \in \mathbb{R}^{nT}$, where each constituent vector $\mathbf{x}_j \in \Sigma_k, j \in I$. argsort $\downarrow(\mathbf{x})$ denotes a function that returns the sorted indices of vector $\mathbf{x}$ in descending order of the magnitude of $\mathbf{x}_i$. The symbol $\odot$ denotes element-wise multiplication of two vectors, as in $\mathbf{z} = \mathbf{x} \odot \mathbf{y}$, where $\mathbf{z}_i = \mathbf{x}_i \mathbf{y}_i$. The operator $\|\mathbf{z}\|_{1,\mathbf{w}} \triangleq \sum_{i=1}^n \mathbf{w}_i \mathbf{z}_i$ is a weighted 1-norm of a vector $\mathbf{z} \in \mathbb{R}^n$, with the weight vector $\mathbf{w} \in \mathbb{R}^n$. A continuous function $\phi : [0, a) \to [0, \infty)$ is said to belong to class $\mathcal{K}$ if it is strictly increasing and $\phi(0) = 0$. Both e^x and $\exp(x)$ are used to represent the exponential function. The symbol $*$ denotes the convolution operator of two vectors; given two vectors $\mathbf{u} \in \mathbb{R}^m$ and $\mathbf{v} \in \mathbb{R}^n$,

The symbol $\mathbb{E}$ denotes the expectation operator of a random variable. $z \sim \mathcal{B}(1,p)$ represents a Bernoulli distributed random variable z, with $\mathbb{E}[z] = p$. Thus, the sum of independent Bernoulli random variables $\{z_1, z_2, \cdots, z_N\}$ with $z_i \sim \mathcal{B}(1,p_i)$ satisfies the Poisson-binomial distribution , and the closed-form expression of the probability density function (pdf) of $\sum_i z_i$ is given by [50]: where with $s_i = -\frac{1-p_i}{p_i}$. The following lemma presents a result that will be useful for subsequent developments in this paper. For any non-decreasing convex function $\Phi : \mathbb{R} \to \mathbb{R}_+$ with $\Phi(0) = 1$, Based on the definition of expectation, Since $\Phi(x) > 0$, then Also, since Φ is non-decreasing and $\Phi(0) = 1$, we have that $\Phi(x) \geqslant 1 \quad \forall x > 0$. Thus,

$\|\mathbf{x}\|_0 \leq k$. A time window containing T samples and ending at index i is denoted by $\mathcal{I} = [i-T+1, i]$. Stacked quantities over this window carry the subscript $\mathcal{I}$.

The expectation and probability operators are denoted by $\mathbb{E}$ and $\Pr$, respectively. The Gaussian distribution with mean $\boldsymbol{\mu}$ and covariance P is $\mathcal{N}(\boldsymbol{\mu}, P)$. The central and noncentral chi-square distributions with ν degrees of freedom and noncentrality parameter λ are denoted by χ^2_ν and $\chi'^2_\nu(\lambda)$. Their cumulative distribution functions are denoted by $F_{\chi^2_\nu}$ and $F_{\chi'^2_\nu(\lambda)}$.

III. System and Threat Model Development

The following model is used to describe the physical process of a CPS: Consider the discrete-time stochastic linear system

$$\begin{aligned}\mathbf{x}_{i+1} &= A\mathbf{x}_i + B\mathbf{u}_i + \mathbf{w}_i,\\ \mathbf{y}_i &= C\mathbf{x}_i + \mathbf{v}_i,\end{aligned} \tag{1}$$

where $\mathbf{x}_i \in \mathbb{R}^n$ is the statevector, $\mathbf{u}_i \in \mathbb{R}^p$ the control input$\mathbf{x}_i \in \mathbb{R}^n$, $\mathbf{u}_i \in \mathbb{R}^p$, and $\mathbf{y}_i \in \mathbb{R}^m$ denote the state, control input, and measurement, respectively. The process noise $\mathbf{w}_i \sim \mathcal{N}(\mathbf{0}, Q)$ and measurement noise $\mathbf{v}_i \sim \mathcal{N}(\mathbf{0}, R)$ are mutually independent white sequences, with $Q \succeq 0$ and $\mathbf{y}_i \in \mathbb{R}^{m_1}$ the sensor measurement. The appropriately dimensioned system matrices A, B and C are assumed to satisfy the following : $R \succ 0$. The initial state is Gaussian and independent of both noise sequences. The following assumptions are imposed:

- A1. The pair (A, B) is controllablestabilizable.
- A2. There exists a positive integer $k_0 \leq m_1$ $k_0 \leq m$ such that the pair $(A, C_{\mathcal{S}})$ is observable for all $\mathcal{S} \subset \{1, 2, \cdots, m_1\}$ $\mathcal{S} \subset \{1, 2, \cdots, m\}$ with $|\mathcal{S}| \geq k_0$.

A3. $\rho\left(A^T\right) > 0$, where the integer T is a specified horizon parameterThe pair $(A, Q^{1/2})$ is stabilizable.

Assumption A2 quantifies the redundancy of the sensing architecture. Specifically, loss of as many as $m - k_0$ sensors does not destroy observability, provided that the remaining measurements are attack-free.

To estimate the system states from sensor measurements, the following receding horizon observer is considered: where

A. Kalman filter

Let $\hat{\mathbf{x}}_{i|i-1}$ and where Thus, Consequently, the following control law is considered: $P_{i|i-1}$ denote the one-step state prediction and its error covariance. With a correctly initialized covariance, the Kalman filter recursion is

$$\hat{\mathbf{x}}_{i|i-1} = A\hat{\mathbf{x}}_{i-1|i-1} + B\mathbf{u}_{i-1}, \tag{2}$$
$$P_{i|i-1} = AP_{i-1|i-1}A^\top + Q, \tag{3}$$
$$\mathbf{r}_i = \mathbf{y}_i - C\hat{\mathbf{x}}_{i|i-1}, \tag{4}$$
$$S_i = CP_{i|i-1}C^\top + R, \tag{5}$$
$$K_i = P_{i|i-1}C^\top S_i^{-1}, \tag{6}$$
$$\hat{\mathbf{x}}_{i|i} = \hat{\mathbf{x}}_{i|i-1} + K_i\mathbf{r}_i, \tag{7}$$
$$P_{i|i} = (I - K_iC)\, P_{i|i-1}. \tag{8}$$

where $\hat{\mathbf{x}}_i \in \mathbb{R}^n$ is the state estimate in (**??**), which is well-known (see [51] for example) to achieve $\lim_{i\to\infty} \|\hat{\mathbf{x}}_i - \mathbf{x}_i\|_2 = 0$Under the nominal hypothesis, the innovation sequence is independent and satisfies $\mathbf{r}_i \sim \mathcal{N}(\mathbf{0}, S_i)$. Under A2–A3, the covariance recursion converges for every measurement subset covered by A2. In the full-sensor case, denote the limiting predicted covariance, gain, and innovation covariance by P^-, K, and the gain matrix K_p is designed to achieve $\rho\,(A + BK_p) \in (0,\ 1)$. Substituting (12) into the dynamics in (1) yields $S = CP^-C^\top + R$, respectively. Hereafter, $\hat{\mathbf{x}}_i \triangleq \hat{\mathbf{x}}_{i|i}$.

B. Chi-square bad-data detector

The normalized innovation squared is

$$g_i \triangleq \mathbf{r}_i^\top S_i^{-1}\mathbf{r}_i \tag{9}$$

which implies that Under the nominal hypothesis $\mathcal{H}_0$, $g_i \sim \chi^2_m$. For a prescribed nominal false-alarm probability $\beta \in (0, 1)$, define

$$\mathsf{BDD}(\mathbf{y}_i) = \begin{cases} 0, & \text{if } g_i \le \tau_{\chi^2}(\beta), \\ 1, & \text{otherwise,} \end{cases} \tag{10}$$

Thus, where $\tau_{\chi^2}(\beta) = F^{-1}_{\chi^2_m}(1-\beta)$. For a window $I = [i - T + 1, i]$, independence of the nominal innovations gives

$$g_I \triangleq \sum_{j\in I} g_j \sim \chi^2_{Tm}. \tag{11}$$

Since $\rho(A + BK) \in (0, 1)$, then $\sum_{i=0}^{\infty} \rho\left((A+BK)^i\right) \le \sum_{i=0}^{\infty} (\rho(A+BK))^i < \infty$[1]. Therefore, the resulting closed-loop dynamics satisfies

$$\sum_{i=0}^{\infty} \|\mathbf{x}_i\|_2 \le \alpha_0 \|\mathbf{x}_0\|_2 + \varphi\left(\sum_{i=0}^{\infty} \|\hat{\mathbf{x}}_i - \mathbf{x}_i\|_2\right)$$

[1]For any $\beta \in (0, 1)$, $\sum_{i=0}^{\infty} \beta^i = \lim_{N\to\infty} \sum_{i=0}^{N} \beta^i = \lim_{N\to\infty} \frac{1-\beta^{N+1}}{1-\beta} = \frac{1}{1-\beta} < \infty$

for a positive constant $\alpha_0 < \infty$ and a class $\mathcal{K}$ functional φ. Since $\hat{\mathbf{x}}_i \to \mathbf{x}_i$, it follows that the origin of the closed-loop dynamics is asymptotically stable [52]. Thus the detector threshold has a direct nominal false-alarm interpretation. Its probability of detection under attack depends on the attack-induced noncentrality parameter developed in Section IV.

Let $\tilde{\mathbf{x}}_i \triangleq \mathbf{x}_i - \hat{\mathbf{x}}_i$, then the closed loop system is given by

C. Closed-loop control

The state estimate is supplied to the certainty-equivalence control law

$$\mathbf{u}_i = K_p\hat{\mathbf{x}}_i, \tag{12}$$

where $\bar{A} = A + BK_p$ and $\bar{B} = BK_p$. Iterating the difference equation above T steps forward yields K_p is selected such that $\rho(A + BK_p) < 1$. Substitution into (1) gives

$$\mathbf{x}_{i+1} = (A + BK_p)\,\mathbf{x}_i + BK_p\,(\hat{\mathbf{x}}_i - \mathbf{x}_i) + \mathbf{w}_i.$$

Consequently, bounded second moments of the estimation error imply bounded second moments of the closed-loop state. In the attack-free case, $\mathbb{E}\|\hat{\mathbf{x}}_i - \mathbf{x}_i\|_2^2 = \mathrm{tr}(P_{i|i})$, and the mean closed-loop dynamics are asymptotically stable [52], [53].

IV. Automated Attack Generation

In this section, a generative model is developed for generating successful FDIAs targeting the model in (1). Including attack signals in the CPS model in (1) yields the adversarial measurement model Consider the attacked measurement model

$$\mathbf{y}_i^a = C\mathbf{x}_i^a + \mathbf{v}_i + \mathbf{e}_i, \tag{13}$$

where $\mathbf{e}_I = \begin{bmatrix} \mathbf{e}_{i-T+1}^\top & \mathbf{e}_{i-T+2}^\top & \cdots & \mathbf{e}_i^\top \end{bmatrix}^\top$ contains the attack vectors within the moving window $[i - T + 1,\ \ i]$. Consequently, substituting (13) into (**??**) yields $\mathbf{e}_i \in \mathbb{R}^m$ is k-sparse. Superscripts a and 0 denote attacked and nominal closed-loop trajectories driven by identical realizations of the process and measurement noises. The analysis considers deterministic attacks that are independent of these noises and begin at the left endpoint of the detection window. The objective of the attack generator is to produce sequences that significantly perturb a critical output while remaining difficult to detect using (10).

A. Attack-induced innovation and output maps

The discriminator construction is developed for the steady-state Kalman gain K. Define

$$\delta\mathbf{x}_i = \mathbf{x}_i^a - \mathbf{x}_i^0, \qquad \delta\hat{\mathbf{x}}_{i|i-1} = \hat{\mathbf{x}}_{i|i-1}^a - \hat{\mathbf{x}}_{i|i-1}^0,$$

Thus, the instantaneous estimation error is given by and let $\boldsymbol{\zeta}_i = \mathrm{col}(\delta\mathbf{x}_i, \delta\hat{\mathbf{x}}_{i|i-1})$. The attack-induced innovation is

$$\delta\mathbf{r}_i = \mathbf{r}_i^a - \mathbf{r}_i^0 = D\boldsymbol{\zeta}_i + \mathbf{e}_i, \qquad D = \begin{bmatrix} C & -C \end{bmatrix}. \tag{14}$$

In general, attackers aim to achieve maximum perturbation of certain critical system output without triggering an alarm from the bad data detection mechanism. For this paper, a residual-based bad data detection scheme is used. Let Under the control

law (12), direct substitution of the KF update gives the augmented recursion

$$\boldsymbol{\zeta}_{i+1} = F_a \boldsymbol{\zeta}_i + G_a \mathbf{e}_i, \tag{15}$$

where $\mathbf{z}$ is given in (**??**), be the residual vector associated with the receding horizon estimator in (**??**). Given a threshold value $\tau \in \mathbb{R}$, the bad data indicator for the windowed measurement history is given by where, with $A_c = A + BK_p$,

$$F_a = \begin{bmatrix} A + BK_pKC & BK_p(I - KC) \\ A_cKC & A_c(I - KC) \end{bmatrix},$$
$$G_a = \begin{bmatrix} BK_pK \\ A_cK \end{bmatrix}. \tag{16}$$

Substituting (**??**) and (13) into (**??**) yields Equations (14)–(16) account for the propagation of earlier attacks through both the plant and the estimator. For a window $\mathcal{I} = [i-T+1, i]$ and zero attack-induced deviation at its beginning, unrolling these equations yields

$$\delta \mathbf{r}_{\mathcal{I}} = \mathcal{H}_r \mathbf{e}_{\mathcal{I}}, \qquad M_2 = \mathrm{blkdiag}(S^{-1/2}, \ldots, S^{-1/2})\mathcal{H}_r, \tag{17}$$

Thus, from the attacker's perspective, it is desirable to keep the quantity $\left\| \left(I - H_0 H_0^{\dagger} \right) \mathbf{e}_I \right\|_2$ as small as possible. where $\mathcal{H}_r$ is block lower triangular, with identity diagonal blocks and subdiagonal block (p, q) equal to $DF_a^{p-q-1}G_a$ for $p > q$.

Next, let $\mathbf{y}_c = C_c\mathbf{x}$ be a vector of critical output targeted by the attacker. We remark that this is not an attempt to identify a priori the particular outputs targeted by an attacker. Rather, we postulate that there are critical functional outputs which the resilient system designer desires to protect against adversarial targeting. Thus, we define FDIAs which target those outputs specifically so that the resulting resilient observer can offer the desired protection. In addition, since most of existing attack generation approaches [4], [18] consider the deviation on state estimates as the attack effectiveness, the associated control law is often omitted in the attack generation problem. However, from the development above, it is necessary to consider the controller since the trajectory of the critical output vector $\mathbf{y}_c$ is also affected by the control law which is also affected by the compromised measurements. Substituting the control law in (12) into the linear model in (1) yields the closed loop system Let $\mathbf{y}_{c,i} = C_c\mathbf{x}_i$ denote the critical output. The corresponding stacked perturbation satisfies

$$\delta \mathbf{y}_{c,\mathcal{I}} = \mathcal{H}_c \mathbf{e}_{\mathcal{I}} \triangleq M_1 \mathbf{e}_{\mathcal{I}}, \tag{18}$$

where $\bar{A} = A + BK_p$ and $\bar{B} = BK_p$. Therefore, Thus, from the attacker perspective, it is desirable to keep the quantity $\left\| C_c \bar{B} A^T H_0^{\dagger} \mathbf{e}_I \right\|_2$ as big as possible. $\mathcal{H}_c$ is obtained by unrolling (15) with output matrix $\begin{bmatrix} C_c & 0 \end{bmatrix}$. Thus M_1 captures the complete plant–controller–estimator response, whereas M_2 captures the complete normalized innovation response.

We now have all the ingredients to formally define and synthesize the generative model for successful FDIAs in CPSs.

B. Successful attacks and generative model

Under the deterministic, noise-independent, window-initialized attack model stated above, the windowed statistic has the noncentral chi-square distribution

$$g_{\mathcal{I}}^a \sim \chi'^2_{Tm}(\lambda_{\mathcal{I}}), \qquad \lambda_{\mathcal{I}} = \|M_2 \mathbf{e}_{\mathcal{I}}\|_2^2. \tag{19}$$

Accordingly, $\lambda_{\mathcal{I}}$ determines the detection probability

$$P_D(\mathbf{e}_{\mathcal{I}}) = 1 - F_{\chi'^2_{Tm}(\lambda_{\mathcal{I}})}\left(\tau_{\chi^2,\mathcal{I}}(\beta)\right), \tag{20}$$

where $\tau_{\chi^2,\mathcal{I}}(\beta) = F^{-1}_{\chi^2_{Tm}}(1-\beta)$.

Consider the CPS in (1). Given a positive integer $k < m$, the attack sequence $\mathbf{e}_I \in \Sigma_k \subset \mathbb{R}^{(T+1)m}$, is said to be (ϵ, α)

Definition 1 ((α, γ)-successful FDIA)**.** Define the per-time sparsity set

$$\Sigma_k^{(T)} = \{\mathbf{e}_{\mathcal{I}} : \|\mathbf{e}_j\|_0 \le k, \ j \in \mathcal{I}\}.$$

An attack sequence $\mathbf{e}_{\mathcal{I}} \in \Sigma_k^{(T)}$ is *(α, γ)-successful if*

$$\|M_1 \mathbf{e}_{\mathcal{I}}\|_2 \ge \alpha, \qquad P_D(\mathbf{e}_{\mathcal{I}}) \le \gamma, \tag{21}$$

for some $\alpha, \epsilon \in \mathbb{R}_+$. where $\alpha > 0$ is the required critical-output perturbation and $\gamma \in (\beta, 1)$ is the maximum admissible detection probability.

The first criterion above requires the effectiveness to be at least α, while the second criterion guarantees the stealthiness of the attack. Earlier researches formulated the attack generation problem as an effectiveness maximization problem with the stealthiness constraint [4], [17]. However, this limits the development of attack generators and its application range. Thus, feasibility-based definitions, as above, are currently favored by researchers [54]. In the above definition, k quantifies the attack sparsity level per time. Specifically, it is the maximum number of attacks allowed at each time index. Examples of such successful FDIAs can be found in literature; a physics-based FDIA generator capable of fooling ℓ_2 observer, such as Kalman filer, can be found in [17] and an optimization-based FDIA generator capable of fooling stronger ℓ_1 observer was proposed in [18]. However, these FDIAs are either computationally too expensive or too conservative to implement on fast real-time systems. In order to circumvent these limitations, we drew inspiration from the well-known generative adversarial networks (GAN) [48] to build generative models for the elements of the set specified by inequalities in (21). The Definition 1 defines a target set of successful attack signals of the form: Because (20) is monotone in $\lambda_{\mathcal{I}}$, the second condition is equivalently represented by

$$\|M_2 \mathbf{e}_{\mathcal{I}}\|_2^2 \le \lambda_{\max}(\beta, \gamma), \tag{22}$$

Thus, the goal of the generative model is to learn how to generate elements of that set. Given a set $\mathcal{S} \subset \mathbb{R}^n$ and a prior distribution $P_z(\mathbf{z})$, a generative model is a mapping of the form $f : \mathbf{z} \mapsto \mathbb{R}^n$ such that $f(\mathbf{z}) \in \mathcal{S}$ with high probability. We consider a generative model of the form $G(\mathbf{z}; \boldsymbol{\theta}_g)$, where $\boldsymbol{\theta}_g$ is a tunable parameter vector that will be prescribed to maximize the likelihood of generating elements in $\mathcal{S}$. Specifically, G is taken as a neural network whose input is a random number $\mathbf{z}$ sampled from a prior distribution P_z, with weights and biases contained in $\boldsymbol{\theta}_g$. Next, similarly to the GAN framework, where $\lambda_{\max}$ is the unique noncentrality level that yields $P_D = \gamma$. This replaces an ad hoc residual threshold with a detector-calibrated stealthiness constraint.

The feasible attack set is therefore

$$\mathcal{S}(\alpha, \gamma) = \{\mathbf{e} \in \Sigma_k^{(T)} : \|M_1 \mathbf{e}\|_2 \ge \alpha, \ \|M_2 \mathbf{e}\|_2^2 \le \lambda_{\max}(\beta, \gamma)\}. \tag{23}$$

We approximate this set using a neural generator $G(\mathbf{z}; \boldsymbol{\theta}_g)$is trained using a fixed discriminator defined by the boundary functions $\|M_1 \mathbf{e}\|_2$, with $\mathbf{z} \sim P_z$. Let $\Pi_{k,T}$ retain the k largest-magnitude entries in each time block and set all other entries to zero. The generated attack is defined as

$$\mathbf{e}_{\mathcal{I}} = \Pi_{k,T}(G(\mathbf{z}; \boldsymbol{\theta}_g)), \tag{24}$$

which enforces membership in $\Sigma_k^{(T)}$. The plant and filter maps M_1 and $\|M_2 \mathbf{e}\|_2$ of $\mathcal{S}$. M_2 act as fixed physics-based discriminators, as illustrated in Fig. 1.

Physics-based generative model and training process (α is the lower-bound constraint for attack effectiveness while ϵ is the upper-bound constraint for attack stealthiness)

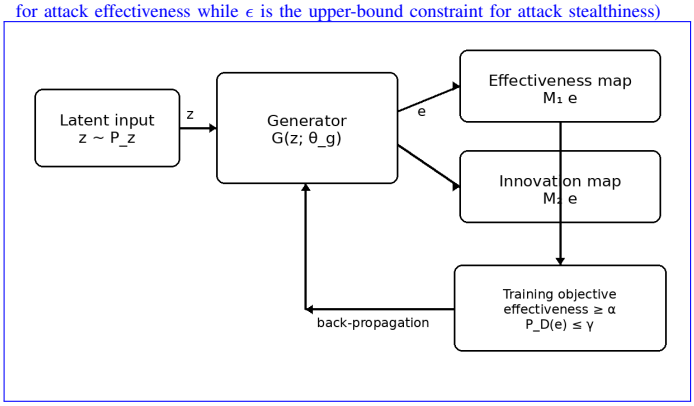


Fig. 1. Physics-guided attack generation. The discriminator M_1 evaluates critical-output effectiveness, whereas M_2 determines the noncentrality parameter of the windowed chi-square detector.

The training process is shown in Fig. 1. The output of the generator is passed, through the discriminator boundary functions, to a loss function designed such that the weights of the generator network are trained to ensure that the outputs of the discriminators satisfy the constraints. Since the constraints are threshold-based, we use an indicator-like loss function:

Consequently, the generator is trained by solving the unconstrained optimization problem

The generator is trained using

$$J(\mathbf{e}) = \mathrm{ReLU}(\alpha - \|M_1\mathbf{e}\|_2) + \mathrm{ReLU}\big(\|M_2\mathbf{e}\|_2^2 - \lambda_{\max}(\beta,\gamma)\big)\,, \tag{25}$$

$$\boldsymbol{\theta}_g^\star = \arg\min_{\boldsymbol{\theta}_g} \mathbb{E}_{\mathbf{z}\sim P_z}\left[J(\Pi_{k,T}(G(\mathbf{z};\boldsymbol{\theta}_g)))\right]. \tag{26}$$

via back propagation.

Please update Fig. 1 to reflect the description above. Specifically, replace the Loss Fun with the optimization problem above. Change "Loss Fun" to "training"

The resulting labeled samples are used to train the channel-wise localization model in Section V.

V. Detection and Localization

A little motivation to connect this section to the previous one. We already have (Model Development) which built physics-based model for the discriminator in GM training for (Automated Attack Generation). How does this section fit into that narrative? Use one or two simple very clear sentences for this.

Through an automated attack generation by a well-trained generative model, an adversarial training dataset could be generated and used to improve the precision of the supervised learning-based attack detection and localization algorithm. In this section, a detection and localization algorithm is given by training a multi-layer perceptron (MLP) on the generated attack dataset. In addition, by analyzing the uncertainty of the MLP's output, a pruning algorithm is proposed to improve the localization precision further.

The term `attack location` refers to sensor channels (or indices) through which the attack signals are added to the system. Let $\mathcal{T}$ denote the actual attack support, then we define its indicator element-wise as:

Next, the definitions of detection and localization are given: Given a measurement vector $\mathbf{y} \in \mathbb{R}^m$, detection is a function $D : \mathbb{R}^m \to \{0,1\}$ whose output $D(\mathbf{y})$ indicates the existence of attack signal in the input $\mathbf{y}$.

Given a measurement vector $\mathbf{y} \in \mathbb{R}^m$, localization is a function $L : \mathbb{R}^m \to \{0,1\}^m$ whose output indicates detection result for each measurement channel. In particular, it is an estimate $\hat{\mathcal{T}}$ of the actual attack support $\mathcal{T}$. Its indicator $\hat{\mathbf{q}}$ is defined similarly to (27), using $\hat{\mathcal{T}}$

Diagram of Training data generation process for multi-class MLP

This MLP section needs to be redone completely. Please connect with the paper. Currently, it reads like a class note. This is a research article. Do NOT write as if you are teaching the reader what MLP is. But write as we are applying MLP to the problem. So, the equations, language and training description should connect with the rest of the paper. Also eqn (19) seems incomplete. What is it? Please put in perspective. Specifically answer the following questions in your write up:

- What do we want to achieve and why is MLP a good choice?
- What is the equation that describe the input-output relationship of the particular MLP for this paper? – Make the equation connect with the rest of the paper. For instance, it seems f needs to be $\hat{\mathbf{q}}$?
- How is the training data generated. You need to describe this very simply and clearly.
- How will the trained MLP be used within the resilient algorithm suite?

Multilayer Perceptron (MLP) is used to detect FDIAs. MLP is a special feed-forward neural networks where the information travels in one direction through all layers. It consists of several layers of nodes: an input layer, a hidden layer, and an output layer.MLPs are capable identifying high dimension non-linearly separable features. The output for a single perceptron in an MLP is given by: where φ is the non-linear activation function, f is the estimated output from the activation function, w is the weight, s_i is the input variable, b is the bias, and n is the input dimension. The activation function is a crucial feature of deep learning algorithms as it determines if and to what extent a neuron output should be considered. For this paper, Rectified linear unit (RELU), Sigmoid, and hyperbolic tangent are used [55]. Cross-entropy loss is utilized to evaluate the detection of the MLP classification and can be calculated as:

As universal function approximators, shown in Cybenko's theorem [56], MLPs

The attacks generated in Section IV are used to train a channel-wise classifier. The classifier serves two distinct purposes. First, it identifies measurement channels that are likely to be compromised. Second, its calibrated channel probabilities are used to describe mathematical models via a regression scheme. Attack detection and localization is a multi-line binary classification task. With adequate training dataset generated by the automated attack generator, MLP's advantages of being a supervised learning network could be fully utilized. construct a sensor subset for resilient state estimation.

The MLP used in this paper is a three-layer neural network containing two *LeakyReLU*

A. Channel-wise localization

Let $\mathcal{T} \subseteq \{1,\ldots,m\}$ denote the attacked sensor set and define the safe-sensor indicator

$$q_j = \begin{cases} 0, & j \in \mathcal{T}, \\ 1, & j \notin \mathcal{T}. \end{cases} \tag{27}$$

The localization model is a vector-valued map $g(\mathbf{y};\boldsymbol{\theta}) \in (0,1)^m$, whose jth component estimates the probability that sensor j is safe. In the numerical study, g is a four-layer fully connected network with three ReLU hidden layers and one *Sigmoid* output layer, and the final classification outputs are 0 (for attack) if the corresponding MLP output is less than 0.5, and 1 (for no attack) otherwise. The output layer's size is the same as the input size and equal to the number of attacks. Consequently, the MLP is given by where ϕ is the activation function of the output layer and $\boldsymbol{\theta}$ is a vector of tunable weights and biases. Then, the localization output is given by a sigmoid output layer. Its binary decision is

$$\hat{q}_j = \mathbb{I}\big\{g_j(\mathbf{y};\boldsymbol{\theta}) \geq \tfrac{1}{2}\big\}\,. \tag{28}$$

Next, a supervised learning algorithm, via back propagation, is used to train the MLP using the Given training pairs $\{(\mathbf{y}^{(i)}, \mathbf{q}^{(i)})\}_{i=1}^{N}$, the parameters are obtained by minimizing the channel-wise binary cross-entropy loss function:

$$J(\boldsymbol{\theta}) = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{m}\big[q_j^{(i)}\log g_j(\mathbf{y}^{(i)};\boldsymbol{\theta}) + (1-q_j^{(i)})\log(1-g_j(\mathbf{y}^{(i)};\boldsymbol{\theta}))\big]. \tag{29}$$

where $(\mathbf{y}^{(i)}, \mathbf{q}^{(i)})$ is the ith training data, and N is the total number of data pointsThe generative model enriches the training set with attacks near the effectiveness and detectability boundaries in (21).

There seems to be a disconnect from the MLP section. Start by motivating the need for uncertainty modeling. We need a brief discussion of the purpose of the pruning algorithm. Why is uncertainty model needed? What will the uncertainty model be used for? – to develop the pruning algorithm Continue with the rest of the development and end with a very clear and concise description of how the pruning algorithm will be used in the resilient estimation. Since it is impossible to completely describe the pruning process here, refer the reader to our earlier paper on the topic and make sure what you have here is a good SUMMARY. Omit details like Lemma V.1. I know you used the lemma because of $\mathbf{r}(k)$ in

B. Calibrated safe-set pruning

Let $\hat{\mathcal{T}}^c = \{j : \hat{q}_j = 1\}$ be the l_η equation. You can just use where clause to state $\mathbf{r}(k)$ instead of the Lemma. The Lemma will make it difficult to follow the paper because there is no proof and readers cannot imagine the proof based on this paper - a lot of missing ingredients. Since the MLP-based attack detection and localization method described above is a multi-line binary classification, the agreement between the localization result $\hat{\mathbf{q}}$ and the actual attack support $\mathbf{q}$ can be described by a Bernoulli distributed agreement variable $\epsilon_i \sim \mathcal{B}(1, \mathbf{p}_i)$ given by set declared safe by the classifier, and let $\mathcal{G}$ denote the sigma-field generated by the complete classifier output and the resulting ordering information. A held-out calibration set is used to estimate, for each candidate $j \in \hat{\mathcal{T}}^c$, the conditional safety probability

$$p_j = \Pr\{q_j = 1 \mid \mathcal{G}\}, \tag{30}$$

where $\mathbf{p}_i = \mathbb{E}[\epsilon_i] = \Pr\{\epsilon_i = 1\}$ is the confidence of the localization algorithm at i channel, which is often obtained by the true rate in *Receiver Operator Characteristic* (ROC) of the algorithmwhere calibration may be implemented by conditioning on score bins and other features included in $\mathcal{G}$. The following analysis assumes that these conditional probabilities are calibrated and that the channel safety events are conditionally independent given $\mathcal{G}$. These assumptions can be relaxed by replacing the product below with a joint lower confidence bound.

$\mathbf{q}, \hat{\mathbf{q}}$ are indicator of unknown actual support prior $\mathcal{T}$ and predicted support prior $\hat{\mathcal{T}}$ by localization algorithm respectively, defined element-wise as:

Consequently, the precision of the resulting support prior indicator $\hat{\mathbf{q}}$ is defined as follows: Given an estimate $\hat{\mathbf{q}} \in \{0,1\}^N$ of an unknown attack support indicator $\mathbf{q} \in \{0,1\}^N$, PPV is the proportion of $\mathbf{q}$ that is correctly identified in $\hat{\mathbf{q}}$. It is given by

([44]) Given mutually independent Bernoulli random variables $\epsilon_i \sim \mathcal{B}(1, \mathbf{p}_i)$, $i = 1, \cdots, Tm$, then where, Order the declared-safe indices so that $p_{j_1} \geq p_{j_2} \geq \cdots \geq p_{j_{|\hat{\mathcal{T}}^c|}}$. For a prescribed reliability $\eta \in (0,1)$, define

$$l_\eta = \max\left\{l \in \{0,\ldots,|\hat{\mathcal{T}}^c|\} : \prod_{r=1}^{l} p_{j_r} \geq \eta\right\}, \tag{31}$$

with $\beta = \prod_{i=1}^{Tm} \mathbf{p}_i, \quad \mathbf{s}_i = -\dfrac{1-\mathbf{p}_i}{\mathbf{p}_i}$

Next, with the attack localization result $\hat{\mathcal{T}}$, the estimated support of safe nodes can be represented as $\hat{\mathcal{T}}^c = \{j_1, j_2, \cdots, j_{|\hat{\mathcal{T}}^c|}\}$. A *Pruning Algorithm* is proposed to prune the most probable erroneous localization results in $\hat{\mathcal{T}}^c$empty product equal to one, and select

$$\hat{\mathcal{T}}_\eta^c = \{j_1, \ldots, j_{l_\eta}\}. \tag{32}$$

This construction differs from a count-based confidence bound: it certifies the particular prefix supplied to the estimator.

The first step is to obtain a trust length $l_\eta (\leq Tm)$ based on the confidence of the localization results $\mathbf{p}$. Given a reliability level $\eta \in (0,1)$, the trust length l_η is an integer defined as the maximum number of safe nodes correctly localized in $\hat{\mathcal{T}}^c$ with a probability of

Proposition 1 (Safe-set precision)**.** Under the calibration and conditional-independence assumptions,

$$\Pr\{\hat{\mathcal{T}}_\eta^c \cap \mathcal{T} = \varnothing \mid \mathcal{G}\} = \prod_{r=1}^{l_\eta} p_{j_r} \geq \eta. \tag{33}$$

Equivalently, the safe-set positive predictive value

$$\mathrm{PPV}_{\mathrm{safe}} = \frac{|\hat{\mathcal{T}}_\eta^c \cap \mathcal{T}^c|}{|\hat{\mathcal{T}}_\eta^c|} \tag{34}$$

equals one with probability *at least* η *; Let where* $s_{j_i} = -\frac{1-\mathbf{p}_i}{\mathbf{p}_i}$. *Then,* $Pr\left\{\sum_{i\in\hat{\mathcal{T}}^c}\epsilon_i = k\right\} = \bar{\mathbf{r}}_k$. *Thus* (**??**) *becomes*[1]

$$\begin{aligned} l_\eta &= max\left\{k \middle| Pr\left\{\sum_{i\in\hat{\mathcal{T}}^c}\epsilon_i \leq k\right\} \leq 1-\eta\right\} \\ &= max\left\{k \middle| \sum_{i=0}^{k}\bar{\mathbf{r}}_i \leq 1-\eta\right\}. \end{aligned} \tag{35}$$

whenever $l_\eta > 0$.

Proof. The selected set is clean precisely when all l_η selected channels are safe. Conditional independence given $\mathcal{G}$ and (30) give the product in (33); the inequality follows from (35). Taking expectations with respect to $\mathcal{G}$ also gives the unconditional bound $\Pr\{\hat{\mathcal{T}}_\eta^c \cap \mathcal{T} = \varnothing\} \geq \eta$. □

The second step is to obtain a *Pruned Support Prior* $\hat{\mathcal{T}}_\eta^c$ of size l_η. This is done through the robust extraction: where $\{\cdot\}_1^{l_\eta}$ denotes an index extraction from the first elements to l_η elements. To obtain a pruned support prior consistent with the model assumptions in Section III, its size should satisfy $l_\eta \geq k_0$, where

[1] $\Pr\left\{\sum_{i\in\hat{\mathcal{T}}^c}\epsilon_i \leq k\right\} \leq 1-\eta$ is equivalent to $\Pr\left\{\sum_{i\in\hat{\mathcal{T}}^c}\epsilon_i = 0\right\} + \Pr\left\{\sum_{i\in\hat{\mathcal{T}}^c}\epsilon_i = 1\right\} + \cdots \Pr\left\{\sum_{i\in\hat{\mathcal{T}}^c}\epsilon_i = k\right\} \leq 1-\eta$ since k could only be integer.

C. Sparsity–observability–precision relation

The next result identifies the admissible reliability levels for which pruning preserves observability.

Proposition 2 (Precision and observability)**.** Suppose $\hat{\mathcal{T}}^c$ contains at least k_0 *is a lower bound of* $\hat{\mathcal{T}}_\eta^c$ *such that* $(A, C_{\hat{\mathcal{T}}_\eta^c})$ *is observable. This can be guaranteed by the proper choice of the reliability level* η*, as given in the following theorem. Given a measurement redundancy preservation parameter* $k_0 > 0$*, the size of pruned support prior satisfies* $l_\eta \geq k_0$ *if* candidates. The pruned set is observable under A2 and is clean with probability at least η if

$$\eta \leq \eta_{\text{obs}} \triangleq \prod_{r=1}^{k_0} p_{j_r}. \tag{36}$$

where $\bar{\mathbf{r}}$ *is given by* (**??**)*, and* $\hat{\mathcal{T}}^c$ *is the estimated support of safe nodes from the attack localization algorithm.* Moreover, l_η is nonincreasing in η.

Proof. According to (**??**), a sufficient condition for $l_\eta \geq k_0$ is which is equivalent to: and According to Lemma **??**, let $z = k_0 - \sum_{i\in\hat{\mathcal{T}}^c} \epsilon_i$, and take $\Phi(z) = \exp(z)$ yields a sufficient condition for (**??**) as Since $\Pr\left\{\sum_{i\in\hat{\mathcal{T}}^c} \epsilon_i = k\right\} = \bar{\mathbf{r}}_k$, it follows that Combining (**??**) and (**??**) yields Finally, an upper bound on Condition (36) makes $l = k_0$ feasible in (35); hence $l_\eta \geq k_0$, and A2 implies observability. Cleanliness follows from Proposition 1. Finally, increasing η is obtained as (**??**). can only remove feasible values from the maximization in (35). □

From (**??**), it is clear that, necessarily, the upper bound should be strictly positive. This yields the following pruning requirement on the localization performance parameters:

Attack sparsity imposes a separate physical limit. Let $k = |\mathcal{T}|$ denote the realized number of attacked sensors. Exactly $m - k$ sensors are genuinely safe; hence, the existence of a clean observable subset requires

$$m - k \geq k_0, \qquad \text{or equivalently} \qquad k \leq m - k_0. \tag{37}$$

If this holds, according to the assumption $A2$, with at least k_0 safe nodes remaining in the pruned support prior $\hat{\mathcal{T}}_\eta^c$, $(A, C_{\hat{\mathcal{T}}_\eta^c})$ will be observable. Thus the asymptotic convergence of the estimation error is guaranteed. False-safe declarations cannot relax (37); they only reduce the probability that the selected set is clean. Thus, (37) quantifies the sparsity–observability limit, while (36) quantifies the precision required to realize that limit using a learned localization model.

According to (**??**), it is clear that the preserved number of measurements k_0 through pruning has a upper bound $\sum_{i=0}^{|\hat{\mathcal{T}}^c|} \left(i - log\mathbf{r}_{\hat{\mathcal{T}}^c}(i+1)\right)$ which depends mainly on the attack localization result $\hat{\mathcal{T}}^c$. If $\hat{\mathcal{T}}^c$ covers more exact safe nodes in $\mathcal{T}^c$, then the upper bound is bigger. According to the assumption $A2$, with at least k_0 safe nodes remained in pruned support prior $\hat{\mathcal{T}}_\eta^c$, $(A, C_{\hat{\mathcal{T}}_\eta^c})$ is observable.

Remark 1. The chi-square detector and the localization network quantify different events. Equation (20) describes the probability of a global alarm, whereas (33) describes the probability that the particular sensor subset supplied to the estimator is uncontaminated. Both quantities enter the resilient-estimation analysis, but they should not be conflated.

For pragmatic reasons, it is important to ensure that $l_\eta > 0$ in (35). This is guaranteed if η is chosen such that at least one node is selected into the pruned set. Formally, this condition is given by:

Moreover, it was shown in [18] that the precision PPV_η of the pruned support prior $\hat{\mathcal{T}}_\eta^c$ given by (32) achieves This indicates that, with the probability of at least η, the measurement nodes contained in $\hat{\mathcal{T}}_\eta^c$ are all safe, and $(A, C_{\hat{\mathcal{T}}_\eta^c})$ is observable according to Remark **??**. Thus, with the probability of at least η, exact state estimation can be achieved by using ℓ_2 observer designed based on $(A, C_{\hat{\mathcal{T}}_\eta^c})$. Finally, Algorithm 1 summarizes the attack detection and localization using automated attack generation and pruning operation.

Algorithm 1: Attack detection and localization with automated attack generation and calibrated safe-set pruning

. **Offline:** **Dataset generation and training.**

1) Given time horizon T, prepare M_1, M_2 matrices for physics-based discriminators ← (18), Construct M_1 and M_2 from (18)–(17).
2) Given the number of attacks $|\mathcal{T}|$, randomize attack support $\mathcal{T}$, and obtain its indicator $\mathbf{q}$ ← (27).
3) Given attack indexes α, ϵ, train the generative model (GM) by solving the optimization problem in Generate labeled attacks by solving (26).
4) Generate attack dataset $\{Y\}$ using the trained GM. Evaluate generated attack dataset via simulation.
5) Return the effective subset of attack dataset as the training dataset $\{Y, \mathbf{q}\}$ Train the channel-wise classifier using (29).
6) Train the MLP in (**??**) using the generated dataset $\{Y, \mathbf{q}\}$ via backpropagation Calibrate the conditional safety probabilities in (30).

. **Online:** **Localization and pruning**

1) Input real-time measurements $\mathbf{y}$, obtain the classification result $\hat{\mathbf{q}}$ ← (**??**), (28); Evaluate $g(\mathbf{y}; \boldsymbol{\theta})$ and form $\hat{\mathcal{T}}^c$.
2) Obtain the pruned support prior $\hat{\mathcal{T}}_\eta^c(\hat{\mathbf{q}}_\eta)$ ← (35) , Sort the candidates by calibrated safety probability.
3) Compute l_η from (35) and return $\hat{\mathcal{T}}_\eta^c$ from (32).

Output: $\hat{\mathcal{T}}_\eta^c$ (pruned support prior)

VI. Resilient Estimator

In this section, the ℓ_2 observer in (**??**) is redesigned based on the pruned support generated by the Algorithm 1 in order to maintain secure state estimation subject to FDIA. The scheme of the resilient ℓ_2 observer is shown in Fig. 2. The attack detection and localization algorithm identifies the locations of attacks and return an estimate of the support of safe nodes $\hat{\mathcal{T}}^c$. Then the pruning algorithm is used to downselect to the safest subset of $\hat{\mathcal{T}}^c$. Finally, the downselected subset of measurements $\mathbf{y}_{\hat{\mathcal{T}}_\eta^c}$ is used in the ℓ_2 observer The pruned set $\hat{\mathcal{T}}_\eta^c$ is used to define a reduced-order measurement update. To preserve the nominal chi-square calibration, the set employed at time i is determined from measurements available no later than time $i-1$. The asymptotic analysis below is restricted to a subset $\mathcal{S}$ that remains fixed over the estimation horizon. Time-varying selection requires a separate analysis of the switched Riccati recursion.

Given the pruned support $\hat{\mathcal{T}}_\eta^c$ within the moving window $[i - T + 1, i]$, the resilient ℓ_2 observer is given by :

A. Subset Kalman filter

For a selected set $\mathcal{S} \subseteq \{1, \ldots, m\}$, define

$$\mathbf{y}_{\mathcal{S},i} = C_{\mathcal{S}}\mathbf{x}_i + \mathbf{v}_{\mathcal{S},i} + \mathbf{e}_{\mathcal{S},i}. \tag{38}$$

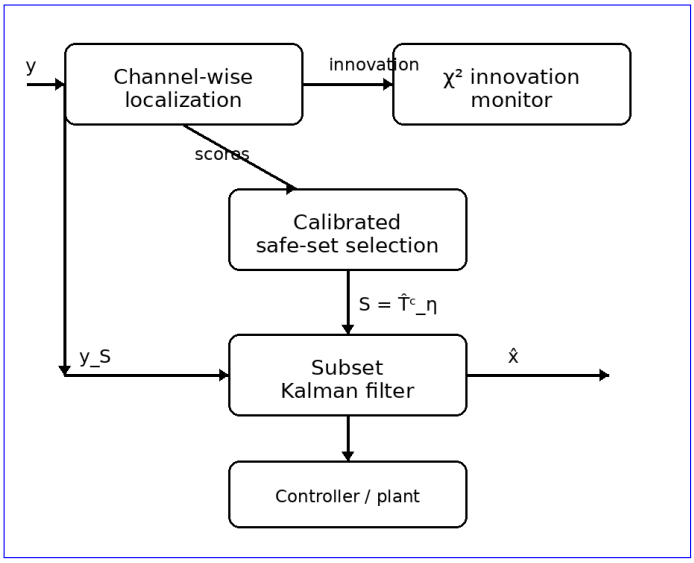


Fig. 2. Schematic diagram of resilient ℓ_2 observerResilient estimation architecture. The localization and pruning stages select a sensor subset, and both the Kalman update and chi-square test are formed using that subset.

where $H_{0,\hat{\mathcal{T}}_\eta^c}$, $H_{1,\hat{\mathcal{T}}_\eta^c}$ are defined similarly to (??) using $C_{\hat{\mathcal{T}}_\eta^c}$ in place of C, and $\mathbf{y}_{\hat{\mathcal{T}}_\eta^c}$ is part of the measurements indexed in $\hat{\mathcal{T}}_\eta^c$. According to (33), $\mathbf{y}_{\hat{\mathcal{T}}_\eta^c}$ contains only clean measurements with a probability of at least η. The next result gives the error bound of the resulting resilient estimatorThe resilient filter applies (2)–(8) with (C, R) replaced by $(C_{\mathcal{S}}, R_{\mathcal{S}})$. Its innovation statistic is

$$g_i^{\mathcal{S}} = (\mathbf{r}_i^{\mathcal{S}})^\top (S_i^{\mathcal{S}})^{-1} \mathbf{r}_i^{\mathcal{S}}. \tag{39}$$

When $\mathcal{S}$ is selected independently of the current innovation and is attack-free, $g_i^{\mathcal{S}} \sim \chi^2_{|\mathcal{S}|}$. Hence, the threshold must be recomputed using $|\mathcal{S}|$ degrees of freedom; a window of T independent subset innovations has $T|\mathcal{S}|$ degrees of freedom.

For a fixed observable subset $\mathcal{S}$, let $P_\infty(\mathcal{S})$ denote the stabilizing filtered solution of the corresponding Riccati equation. Since $Q \succeq 0$, this covariance is positive semidefinite. Additional independent measurements cannot increase the optimal filtering covariance. Therefore, for nested observable sets $\mathcal{S}_1 \subseteq \mathcal{S}_2$,

$$P_\infty(\mathcal{S}_2) \preceq P_\infty(\mathcal{S}_1). \tag{40}$$

Theorem VI.1 (Conditional resilient-estimation performance)**.** *Given a pruned support $\hat{\mathcal{T}}_\eta^c$ from (32). Suppose η satisfies (??) with the underlying attack localization algorithm satisfying the performance criterion in (??). Then, with the probability of at least η, the resilient estimator in (??) achieves the same error performance as the attack-free estimates in (??). Namely,*

$$\|\hat{\mathbf{x}}_i^{'} - \mathbf{x}_i^\star\|_2 = \|\hat{\mathbf{x}}_i - \mathbf{x}_i^\star\|_2 \tag{41}$$

where $\hat{\mathbf{x}}_i^{'}$ is the state estimate obtained using (??), $\hat{\mathbf{x}}_i$ is the attack-free estimate in (??) respectively, and $\mathbf{x}^\star$ is the actual state vector. We show that $\|\hat{\mathbf{x}}_i^{'} - \hat{\mathbf{x}}_i\|_2 = 0$ with probability of at least η. Since $\rho(A^T) > 0$, using the relationship $\hat{\mathbf{x}}_i = A^T\mathbf{z} + F\mathbf{u}_{I-1}$ and $\hat{\mathbf{x}}_i^{'} = A^T\mathbf{z}^{'} + F\mathbf{u}_{I-1}$, it is seen that $\|\hat{\mathbf{x}}_i^{'} - \hat{\mathbf{x}}_i\|_2 = 0$ is equivalent to $\|\mathbf{z}^{'} - \mathbf{z}\|_2 = 0$, where $\mathbf{z}$ and $\mathbf{z}^{'}$ are given by Let $\mathcal{S} = \hat{\mathcal{T}}_\eta^c$ be fixed over the estimation horizon and suppose $\eta \le \eta_{\text{obs}}$ in (36). Conditional on the event

$$\mathcal{E}_{\mathcal{S}} = \{\mathcal{S} \cap \mathcal{T} = \varnothing\},$$

the resilient filter is identical to the attack-free Kalman filter based on $\mathcal{S}$, *and respectively. Since the pruned support prior is generated by (32), then (33) holds, which implies that, with the probability of at least η, $\mathbf{e}_{\hat{\mathcal{T}}_\eta^c} = \mathbf{0}$. Thus, (??) becomes*

$$\lim_{i\to\infty} \mathbb{E}\Big[\|\mathbf{x}_i - \hat{\mathbf{x}}_{i|i}^{\mathcal{S}}\|_2^2 \mid \mathcal{E}_{\mathcal{S}}, \mathcal{S}\Big] = \operatorname{tr} P_\infty(\mathcal{S}). \tag{42}$$

Then combining (??) and (??) yields Furthermore,

$$\begin{aligned}\operatorname{tr} P_\infty(\{1,\dots,m\}) &\le \operatorname{tr} P_\infty(\mathcal{S}) \\ &\le \max_{\mathcal{U}\subseteq\mathcal{S},\ |\mathcal{U}|=k_0} \operatorname{tr} P_\infty(\mathcal{U}).\end{aligned} \tag{43}$$

Since η satisfies (??) andthe underlying attack localization algorithm satisfies the performance criterion in (??), then $|\hat{\mathcal{T}}_\eta^c| > k_0$ which implies $(A, C_{\hat{\mathcal{T}}_\eta^c})$ is observableaccording to the assumption A2. Then, based on the definition of $H_{0,\hat{\mathcal{T}}_\eta^c}$ The conditioning event satisfies $\Pr(\mathcal{E}_{\mathcal{S}} \mid \mathcal{G}) \ge \eta$ and, consequently, $\Pr(\mathcal{E}_{\mathcal{S}}) \ge \eta$.

Proof. Propositions 1 and 2 give $\Pr(\mathcal{E}_{\mathcal{S}} \mid \mathcal{G}) \ge \eta$ and H_0, they both have full rank. Therefore, since $k_0 < m_1$, the Moore-Penrose pseudo-inverses can be directly calculated as $H^\dagger_{0,\hat{\mathcal{T}}_\eta^c} = (H^\top_{0,\hat{\mathcal{T}}_\eta^c} H_{0,\hat{\mathcal{T}}_\eta^c})^{-1} H^\top_{0,\hat{\mathcal{T}}_\eta^c}$ and $H_0^\dagger = (H_0^\top H_0)^{-1} H_0^\top$. Then $\|\mathbf{z}^{'} - \mathbf{z}\|_2 = \|(I-I)\mathbf{x}_{i-T}\|_2 = 0$.

By using ℓ_2 observer in (??), the $|\mathcal{S}| \ge k_0$. Assumption A2 then implies that $(A, C_{\mathcal{S}})$ is observable. On $\mathcal{E}_{\mathcal{S}}$, $\mathbf{e}_{\mathcal{S},i} = 0$ throughout the interval, so the subset-filter recursion coincides exactly with its attack-free state estimation error $\|\hat{\mathbf{x}}_i - \mathbf{x}_i^\star\|_2$ converges to zero as i goes infinity. According to (42), $\|\hat{\mathbf{x}}_i - \mathbf{x}_i^\star\|_2$ also goes to zero in the limit.

In this section, a weighted ℓ_1 minimization program is used for resilient observer design whose weight is defined based on the pruned support prior. Consider the CPS in (1) and the corresponding measurement model (??), given *Prune Support Prior* $\hat{\mathcal{T}}_\eta^c$ by pruning algorithm, counterpart. Standard Kalman-filter convergence gives (42). The first inequality in (43) follows because $\mathcal{S}$ is contained in the corresponding moving-horizon weighted ℓ_1 observer is given by: where, for $\omega \in (0,\ \ 1)$, the weight vector $\mathbf{w}(\hat{\mathcal{T}}_j, \omega) \in \mathbb{R}^m$ is given element-wise as full sensor set. For every k_0-element subset $\mathcal{U} \subseteq \mathcal{S}$, (40) gives $P_\infty(\mathcal{S}) \preceq P_\infty(\mathcal{U})$; taking the largest trace proves the second inequality. □

The optimization problem in (??) is equivalent to where, $\mathbf{w}(\hat{\mathcal{T}}, \omega) = \begin{bmatrix} \mathbf{w}(\hat{\mathcal{T}}_i, \omega) \\ \vdots \\ \mathbf{w}(\hat{\mathcal{T}}_{i-T+1}, \omega) \end{bmatrix} \in \mathbb{R}^{Tm}$. theorem is intentionally conditional. No finite unconditional mean-square bound follows from a cleanliness probability alone because the attack magnitude on the complementary event $\mathcal{E}_{\mathcal{S}}^c$ may be unbounded. A trajectory-wide unconditional guarantee therefore requires either an attack-amplitude model or a robust update on the failure event.

Consider the system model in (1) together with the corresponding measurement model (??). Let $\hat{\mathcal{T}}$ be a predicted support prior generated by the multi-class MLP in Algorithm ??, then $\hat{\mathcal{T}}_\eta^c$ is generated by Algorithm ?? with a positive parameter η. If there exists an integer $a \ge \max\{\rho - 1, 1\}$, where $\rho = \frac{|\hat{\mathcal{T}}_\eta|}{k}$, such that any coding matrix F $(FH = 0)$ satisfies RIP condition for all sparse vector $\mathbf{h} \in \mathcal{S}_S^{Tm}$, and the RIP constant satisfies then, with a probability of at least

B. Quantitative performance relation

Equations (37), (36), and (43) provide the desired quantitative relation. First, $k \le m - k_0$ is necessary for any

clean observable subset to exist. Second, the calibrated probabilities determine the largest reliability level η_{obs} for which the learned selector retains at least k_0 sensors. Third, for every admissible η, the estimation error of the weighted ℓ_1 Observer in (??) can be upper bounded as where $\underline{\sigma}_H$ is the smallest singular value of H, and resulting conditional mean-square error is $\operatorname{tr} P_\infty(\mathcal{S})$.

The relation also exposes a design tradeoff. Increasing η generally reduces l_η, thereby increasing the covariance through (40); decreasing η retains more measurements but weakens the certified probability that all retained sensors are safe. A suitable reliability level may therefore be selected by

$$\min_{\eta\in[\eta_{\min},\eta_{\text{obs}}]} \operatorname{tr} P_\infty(\hat{\mathcal{T}}_\eta^c) \quad \text{subject to} \quad \Pr(\mathcal{E}_{\hat{\mathcal{T}}_\eta^c} \mid \mathcal{G}) \geq \eta. \tag{44}$$

Here $\eta_{\min}$ is a designer-specified reliability floor. This formulation connects localization confidence directly to the estimator covariance rather than treating observability as a binary post-processing check.

VII. Simulation

In this section , the corresponding improvement of resiliency due to the automated attack generation is shown in simulation, by implementing Algorithm 1 and This section evaluates the sensing redundancy and channel-localization components of the resilient observer in (??) on a water distribution systemproposed framework using a water-distribution CPS. The available data establish the empirical observability profile and quantify the effect of physics-guided attack data on localization. The scope of the study is limited to these two components; no KF/χ^2 performance claim is inferred from residual-based simulations of the earlier estimator.

Block diagram depiction of a water distribution system under FDIAs (black solid are water pipelines, blue dotted lines are wireless data transmission lines for sensors data and control commands, orange dotted lines are attack injection route)

The water distribution system , shown

A. Water-distribution model

The system in Fig. 3 , contains 10 distributed contains ten operating tanks and 1 one storage tank. The goal control objective is to regulate all operating tanks' water levels around desired values. The magnetic valves $V_1 - V_{10}$ at the inlet of operating tanks are controlled to adjust the water level of water tanks, and the manual valves$M_1 - M_{10}$ can be manipulated, at the demand point, to use the water from the corresponding tanks. The effect of these valves is modelled as the random demands $\mathbf{d}$ in (45). The valves at the storage tank are fixed at a constant opening value. For each operating tank, two sensors-water level meter and pressure sensors (L_1 L_{10}), are installed. The pressure sensors measure the difference of water level between adjoining tanks. The water levels adjustment process is approximated by the LTI model : the operating-tank levels by adjusting the inlet valves. Random consumer demand is modeled as a process disturbance. Level and differential-pressure measurements are transmitted to the control center and may be compromised by additive FDIAs.

The linearized discrete-time model is

$$\begin{aligned} \mathbf{x}_{i+1} &= A\mathbf{x}_i + B\mathbf{u}_i - \mathbf{d}_i, \\ \mathbf{y}_i &= C\mathbf{x}_i + \mathbf{v}_i + \mathbf{e}_i, \end{aligned} \tag{45}$$

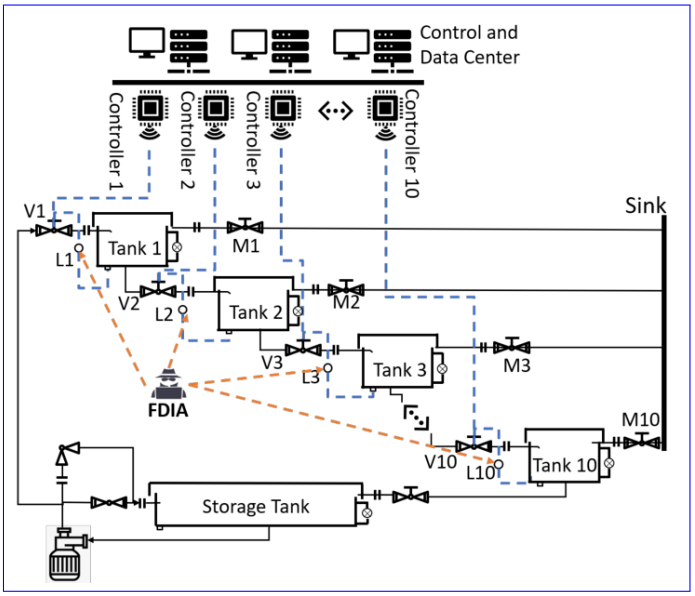


Fig. 3. Water-distribution CPS. Solid lines denote water pipelines, blue dotted lines denote measurement and control communication, and orange dotted lines denote attack routes.

where , $\mathbf{x}, \mathbf{u} \in \mathbb{R}^{10}$, $\mathbf{y}, \mathbf{v} \in \mathbb{R}^{61}$ are vector of water level, control input of magnetic valves, sensor measurements, and measurement noise respectively, $\mathbf{d} \in \mathbb{R}^{10}$ is the vector of random demands, and $\mathbf{e} \in \mathbb{R}^{61}$ is the injected FDIAs. For the sake of clean presentation, interested readers are directed to https://github.com/ZYblend/Resilient_WDS_AutoAttackGen for the values of system parameters $A \in \mathbb{R}^{10\times10}$, $B \in \mathbb{R}^{10\times10}$, $C \in \mathbb{R}^{61\times10}$. Given a desired water level $\mathbf{x}_d = 0.5 * \mathbf{1}_{10}$, the proportional controller below is used; The critical measurement in this scenario is the sum of all water levels of $\mathbf{x}, \mathbf{u} \in \mathbb{R}^{10}$, $\mathbf{d} \in \mathbb{R}^{10}$ is the demand disturbance, and $\mathbf{e}$ is the measurement attack. The critical output is the aggregate level of the operating tanks, thus, the critical measurement matrix is given by $C_c = [1 \quad 1 \quad \cdots \quad 1] \in \mathbb{R}^{1\times10}$. Firstly, a simulation experiment is done to assess the observability of the simulated system in order to obtain the least number of measurements from which the system states are fully observable. A Monte Carlos experiment, with 50000 samples of randomly selected measurement subsets $\mathcal{S}$, is performed to calculated the probability of full observerability of $(A, C_{\mathcal{S}})$ using $|\mathcal{S}|$. Fig 4 shows that full observerability is preserved with $k_0 \geq 42$, corresponding to at most 30% pruned measurements.

Observability of simulated systems. The percentage of observable $(A, C_{\mathcal{S}})$ is calculated as the ratio of the number of observable $(A, C_{\mathcal{S}})$ to the total number of samples (=50000).

Next, we execute different simulation scenarios: automated attack generation, training data preparation with MLP training, and Monte-Carlo simulation for resilient ℓ_2 observer with different attack detection and localization strategiesi.e., $C_c = \mathbf{1}_{10}^\top$. Model-generation and attack-generation code is available at https://github.com/ZYblend/Physics-guided_GAN_Attack_Generation.

B. *Automated attack generation*

A fixed time step of $T_s = 0.01s$ is used and the moving-window size is set as $T = 8T_s$. With different numbers of attacks, the transfer matrices M_1, M_2 are calculated based on the corresponding random attack support. Then, the generator in (26) is trained as shown in Fig. 1. Due to the uncertainty in the discriminator caused by measurement noise and random demands, the threshold values are chosen from experience as $\alpha = 0.1, \epsilon = 0.5$. Fig. ?? shows the effectiveness and detection residual of the generated

attacks during training process compared to the thresholds (α, ϵ). It is seen that the trained generator successfully transforms uniformly distributed inputs to feasible attacks satisfying the conditions in (23). Feasibility of generated attacks during training process. ϵ is the upper bound for the detection residual, α is the lower bound of effectiveness.

The trained generator is then used to generate a dataset $D_{\mathcal{S}}(\alpha, \epsilon)$ of attacks. In order to inject the generated attacks, recursive feasibility [57] should be considered. An online moving-horizon attack injection scheme used here is to solve the following sample-based optimization program at each time step i:

$$\begin{aligned} \underset{\mathbf{e}_i}{\text{Maximize}} \quad & \left\| M_1 \begin{bmatrix} \mathbf{e}^- \\ \mathbf{e}_i \end{bmatrix} \right\|_2 \\ \text{Subject to:} \quad & \left\| M_2 \begin{bmatrix} \mathbf{e}^- \\ \mathbf{e}_i \end{bmatrix} \right\|_2 \leq \epsilon \\ & \mathbf{e}_i \in D_{\mathcal{S}}(\alpha, \epsilon) \end{aligned} \tag{46}$$

where $\mathbf{e}^-$ is the attack history during the time window $I^- = [i - T + 1, i - 1]$. Fig. **??** shows the performance of the generated attacks in system operation through the injection program (46). The left figure shows that the T-horizon cumulative residual under attackshas small deviation from the nominal T-horizon cumulative residual, which means the attacks are hiding in the noise. However, the right figure shows that the critical measurement has obvious deviation from nominal critical measurement after attacks are injected, which means the generated attacks inject targeted biases in the system successfully. Performance of generated attacks injected through 39% measurements nodes of the water distribution system. (Left: T-horizon cumulative residual, Right: critical measurement at each time step.)

Next, the above automated attack generation scheme is trained to generate different types of attacks including different attack supports and different number of attacks. The generated attacks are then used to train the supervised learning-based attack detection and localization algorithm in next subsection.

B. *MLP training*

To show one of the merits of the proposed automated attack generation scheme, two training datasets are prepared: one produced by random additive signals on measurements and the other augments the dataset with attack samples generated by the trained generator. Next,two MLP classifiers are trained on two datasets respectively. MLP1 and MLP2 are both four-layer fully-connected neural network with three "relu" hidden layers of $512, 1024, 512$ neurons respectively. The output layer has 61 neurons and uses "sigmoid" activation function. MLP1 is trained by the first random attack dataset, while MLP2 is trained the second training dataset containing automated generated attacks. The testing F1 score and precision of MLP1 are 85.1% and 99.7% respectively, while MLP2 achieves 88.7% and 99.1% respectively. The testing datasets used is 10% of the corresponding datasets, so the testing datasets for MLP2 and MLP1 are different. Next, we compare the localization precision of the two MLPs on a same testing dataset which is an equal mixture of dataset 1 and dataset 2. As shown in Fig. 7, MLP2 has successfully shifted the localization precision to the right, indicating a better performance overall. However, MLP1's performance is nearly uniform for different localization precision higher than 0.5. This is because it is trained by random attack dataset. The mean precision of MLP1 is 74.37%, and the mean precision of MLP2 is 98.69% on a testing set of successful FDIAs.

A sample-based comparison of the performance of two MLP classifiers on same testing set of successful FDIAs. (MLP1: trained only with random attack generation, MLP2: trained with random attack generation and the proposed automated attack generation, 50000 test cases are used.)

B. Empirical observability profile

For each sensor-set cardinality, $50{,}000$ subsets were sampled uniformly and the rank of the associated observability matrix was evaluated. Figure 4 reports the fraction of sampled subsets for which (A, C_S) was observable.

As shown in Fig. 4, full observability can be guaranteed with certainty if $k_0 \geq 42$ measurements remains after pruning. For this, we evaluate the sufficient condition in

A sample-based comparison of post-pruning observability for MLP1 and MLP2. (50000 test cases are used. The y axis for the top two subplots $Pr(k_j \geq k_0)$ is the sample estimate of the probability of at least k_0 measurements remaining after pruning. The black lines in the bottom subplots show the mean value of k_j that is the number of measurements remaining after pruning, while the red and green dots are the minimum and maximum values of k_j's respectively.

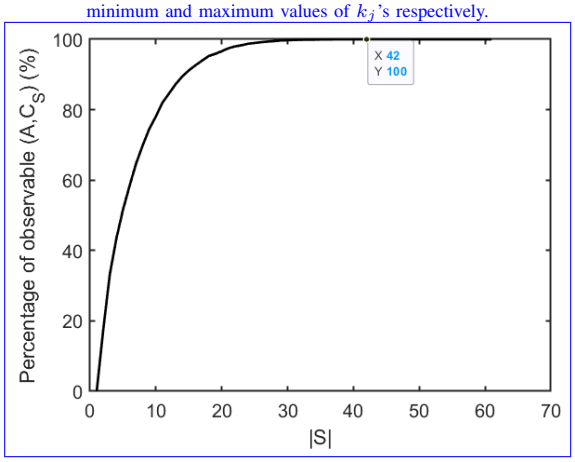


Fig. 4. Empirical frequency of observability as a function of the number of retained measurements (50,000 random subsets per cardinality).

(**??**) on MLP1 and MLP2. First, the two MLPs are run on 10000 cases generated by the trained generative model. From the results, the confidence vectors of the MLPs are estimated by calculating the true rates at all 61 measurement nodes; for node i, $\mathbf{p}_i = \frac{TP_i + TN_i}{P_i + N_i}$. The confidence vectors of MLP1 and MLP2 are Next, a sample based experiment is performed. For each support size $|\hat{\mathcal{T}}_j^c| \in \{40, 41, \ldots, 61\}$, we generated $\min\left\{{}^{61}C_{|\hat{\mathcal{T}}_j^c|}, 50000\right\}$ unique supports $\hat{\mathcal{T}}^c$, and evaluated the right hand side of (**??**) for each unique support as follows: where the Poisson-binomial probability density vector $\bar{\mathbf{r}}_i$ is calculated, using the confidence vectors of the corresponding MLP, based on (**??**). For each $|\hat{\mathcal{T}}_j^c| \in \{40, 41, \ldots, 61\}$, and each MLP, the sample probability of the event $k_j \geq k_0$ and the expected value of k_j are shown in Fig. **??**. The results show that MLP2 has a bigger chance of preserving post-pruning observability. Indeed, according to the figure, post-pruning observability cannot be guaranteed when MLP1 is used. In addition, from the bottom plots in Fig. **??**, it is seen that MLP2 reduces the uncertainty on the post-pruning observability compared to MLP1. By considering the sample-based analysis in Fig. 7 and Fig. **??** together, it is seen that MLP performs better when trained with data generated by the proposed automated attack generation. All sampled subsets containing at least 42 measurements were observable. This observation motivates the empirical design value $k_0 = 42$; it does not constitute an exhaustive verification of A2. With $m = 61$, the corresponding empirical sparsity guideline is $k \leq 19$. Experiments with more than 19 attacked channels lie outside the sampled operating regime.

Next, we present a comparison of

C. Localization performance

Two four-layer MLPs were evaluated. MLP1 and was trained using random additive attacks; MLP2 in the resilient ℓ_2 estimation scheme for the water distribution system to show their effects on the estimator's resiliency.

D. *Resilient ℓ_2 observer*

Finally, the trained MLPs with pruning algorithm are used in a real-time simulation of the water distribution system. The resilient ℓ_2 observer in (**??**) is included to perform the

state estimation . The goal of this simulation is to estimate the probability of achieving resilient performance, using different attack detection strategies with the ℓ_2 observer. A Monte-Carlo experiment is carried out. The input of the experiment are generated attacks by the trained attack generator . The outputs are: (1) the **probability of sustaining normal operation** (pSNO) estimated as the ratio of the instances where the deviation of critical measurement from the nominal value is below a given threshold to the total number of instances, (2) the **probability of triggering an alarm** (pTA) estimated as the ratio of the instances where the deviation of detection residual from the nominal value is beyond a given thresholdto the total number of instances. Clearly, bigger values of pSNO and pTA indicate better resiliency.

The input attack dataset includes different number of attacks from 1 to 30, and for each number of attacks, we generated 5 datasets corresponding to different attack scenarios. The attack supports were generated randomly. Next, the system simulation process shown in Fig. 2 was carried out with different attack detection strategies: MLP1used the same data augmented with attacks produced by the earlier residual-based physics-guided generator. Thus, this experiment evaluates the general benefit of physics-guided data augmentation, rather than the revised generator in Section IV. Both networks contain three hidden layers with 512, MLP1 with pruning, MLP2, and MLP2 with pruning. During the system simulation, the T-horizon cumulative detection residual $\mathbf{r}_I$ is calculated based on (18), and the critical measurement $\mathbf{y}_{c_{i+1}}$ is also calculated based on (17). Then the performance, in terms of deviation ratios of the residual $\Delta\mathbf{r}$ and the critical measurements $\Delta\mathbf{y}$ respectively, are evaluated using where $\mathbf{r}_d$ is the residual vector in the corresponding attack-free and noise-free case, $\mathbf{y}_{c_d} = C_c\mathbf{x}_d$ is the desired critical measurement vector corresponding to the desired states $\mathbf{x}_d$. Consequently, pTA and pSNO are estimated as pTA $= \frac{1}{N}\left\|\Delta\mathbf{r} \geq \frac{\tau_1|\mathbf{y}_c^\star|}{\max(|\mathbf{y}_c-\mathbf{y}_c^\star|)}\right\|_{\ell_0}$ and pSNO $= \frac{1}{N}\left\|\Delta\mathbf{y} \leq \frac{\tau_2|\mathbf{r}^\star|}{\max(|\mathbf{r}-\mathbf{r}^\star|)}\right\|_{\ell_0}$ respectively, where N is the total number of time instances from the start of attack injection. The thresholds $\tau_1 = 1\%$ 1024, and $\tau_2 = 3\%$ are set based on the responses of the attack-free simulation. 512 ReLU units, followed by an m-dimensional sigmoid layer. The comparison uses a common test set of 50,000 FDIAs.

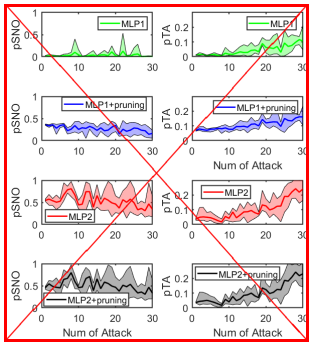


Fig. 5. A comparison of the ℓ_2 observer resiliency with different attack detection strategies, bigger values of pSNO and pTA indicate better resiliency. (The solid lines are mean values and the colored region indicates the spread of quantities)

Fig. 5 shows the result of the experiment and a comparison of pSNO and pTA for different detection strategies. It is seen that MLP2 has better resiliency than MLP1, even better than MLP1 with pruning. This demonstrates the significant improvement in resiliency by including the automated attack generator in the training of MLP. In addition, the mean of localization precision, calculated by (34), of four attack detection strategies are shown in Fig. 6. As discussed in the last subsection, although the pruning algorithm still improve the localization precision of MLP2, the state estimation result is not improved as much due to the loss of observability. This indicates the MLP2 has already approached the best balance between the localization precision and the observability requirement. However, it is noteworthy that MLP2 with pruning has tighter spread of pSNO and pTA values than MLP2. This indicates that the pruning algorithm can reduce performance uncertainty.

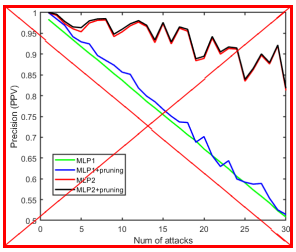


Fig. 6. A comparison of the mean precision of different attack detection strategies in the system simulation versus different number of attacks

Mean value of attack localization precision #(attacks) MLP1 The mean localization precision increased from 74.37% for MLP1 +pruning MLP2 MLP2+pruning 2 0.9150 0.9696 0.9559 0.99933 0.8853 0.9427 0.9282 0.97904 0.8612 0.9182 0.8998 0.95365 0.8356 0.8925 0.8778 0.9381 6 0.8148 0.8651 0.8645 0.9201 7 0.7903 0.8414 0.8402 0.8878 8 0.7672 0.8116 0.8234 0.8823 9 0.7434 0.7905 0.7993 0.8677 10 0.7150 0.7602 0.7719 0.8407 11 0.6905 0.7377 0.7501 0.8296 12 0.6651 0.7091 0.7174 0.7904 13 0.6441 0.6851 0.7011 0.7689 14 0.6167 0.6598 0.6705 0.7496 15 0.5704 0.6349 0.6517 0.738816 0.5704 0.6106 0.6412 0.722317 0.5462 0.5833 0.6135 0.6955 18 0.5251 0.5576 0.5983 0.6751 19 0.5012 0.5305 0.5641 0.6265 20 0.4725 05069 0.5296 0.6192

Finally, a time-domain simulation result is presented in Fig. **??**. The attacks, same as the ones in Fig **??**, are injected through the program (46). Then four resilient estimation schemes are compared. It is seen that the schemes using MLP2 and to 98.69% for MLP2 with pruning reduce the attack effectiveness on the critical measurements (smaller $\Delta\mathbf{y}$), and also reduce the attack stealthiness (bigger $\Delta\mathbf{r}$). This means that, with MLP2 or MLP2 and pruning, it is easier to detect the attacks while, at the same time, their effects are reduced.

A time-domain simulation results of the resilient estimation with four different attack localization strategies when the same 39% measurements are attacked as shown in Fig**??** on the reported test distribution. Together with the empirical value $k_0 = 42$, calibrated classifier scores would determine the admissible reliability η_{obs} through (36). The archived experiment did not record the conditional calibration data or subset Riccati covariances required to evaluate this relation; consequently, no clean-set probability or KF covariance certificate is claimed for these results.

VIII. Conclusion

In this paper, we present an algorithm design for resilient cyber-physical system, the resiliency is significantly improved by including the proposed automated attack generation in the training of attack detection models. Unlike traditional GAN-based FDIA generation, the automated attack generator does not require prior attack samples. This improved resilient estimation framework is a concurrent-learning based approach which contains a data-driven attack generator, an attack detection and localization algorithm, a model-based estimator design, and a pruning algorithm. The attack detection and localization algorithm used in this framework could be any supervised learning-based classification algorithm, not restricted to MLP. This paper developed a resilient estimation architecture that combines a Kalman filter, chi-square innovation monitoring, physics-guided attack generation, and channel-wise localization. The attack generator was formulated using finite-horizon maps that retain the dynamics of both the estimator and

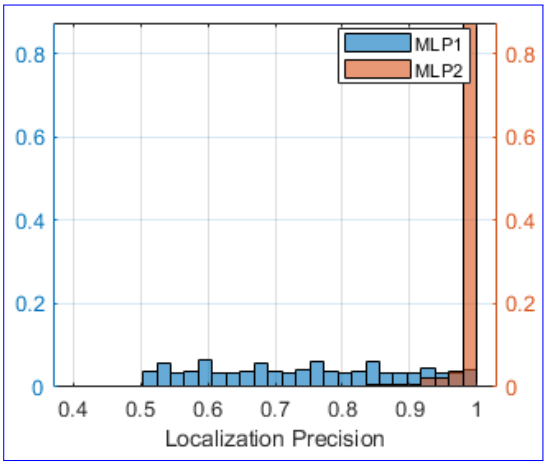


Fig. 7. Distribution of channel-localization precision on a common test set. MLP1 is trained using random attacks; MLP2 is trained using random and physics-guided attacks.

the controlled plant. This construction gives a detector-calibrated measure of stealthiness through the noncentral chi-square distribution, rather than through an unnormalized residual threshold.

There are several open questions to be addressed in future work: The discriminator is constructed based on the knowledge of the system dynamics, although the simulation in this paper suggest that exact knowledge is not necessary for generating feasible attacks. The theoretical convergence proof of the generator with uncertainties on physics-based discriminator is still open. Data-driven discriminators with low computation burden is also a good problem to address. To improve the resiliency further, a weighted $\ell_1-\ell_2$ state estimator could be used, but the theoretical development for bounded estimation error should be given with attack detector and pruning algorithm in the loop, as in Theorem **??**, and Theorem VI.1. Finally, the automated attack generation design could be further studied against the nonlinear systems. The localization output was converted into an estimator input through calibrated safety probabilities. Under explicit calibration and conditional-independence assumptions, the selected subset is attack-free with a prescribed probability. The analysis further showed that attack sparsity limits the existence of a clean observable subset, while the size of the selected subset determines the conditional Kalman-filter covariance through the Riccati equation. The water-distribution case study demonstrated the empirical sensing redundancy and the localization improvement obtained by augmenting the training set with physics-guided attacks. Closed-loop numerical validation of the revised KF/χ^2 architecture is outside the scope of the available data and is not inferred from the earlier residual-based results.

Future work will consider dependent channel-calibration models, time-varying sensor subsets, and unconditional error guarantees obtained by combining probabilistic selection with a robust update on the selection-failure event.

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